Sahaj Bhadja

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Summary

Backend Engineer with 2+ years building **REST APIs**, **SQL**-backed services, and **AWS** deployments in Python and **Java/Spring Boot**, with strong fundamentals, tested code, and malware-detection research experience.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Operating Systems, Object-Oriented Programming, Software Engineering, Data Retrieval

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, CA*

- Designed normalized **PostgreSQL** and Azure SQL schemas with row-level security for 500+ platform users, troubleshooting slow queries and adding indexes that cut dashboard latency from 1.2s to 350ms.
- Built fault-tolerant **backend services** with FastAPI background workers and explicit job states, processing 1K+ AI jobs and 5K+ monthly integration jobs at 99%+ reliability with 3x throughput under concurrency.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*

- Built Flask **REST APIs** with modular Blueprint MVC structure backing a React grading platform, serving 150+ instructors and 10K+ daily requests on AWS while cutting feedback turnaround 60%.
- Implemented secure PDF, DOCX, and ZIP ingestion with concurrent ThreadPoolExecutor batches over **MySQL** on AWS RDS, processing 1K+ files per batch and shipping releases through GitHub Actions CI/CD.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer – SecureSync *India*

- Debugged and optimized large-scale **API integrations** with object-oriented Singleton and Factory patterns, pagination, and caching, improving response times 30% and cutting technical debt 20% across cross-platform microservices.
- Automated 20+ **unit and integration test** suites with PyTest, Mockito, and nightly Jenkins CI/CD across three operating systems, and redesigned AWS Glue and S3 ETL workflows, boosting efficiency 68%.

Cyber Security and Data Mining: Detecting Malicious URL **May 2023 – August 2023**
Data Analyst – Research Fellowship *India*

- Built a **malicious URL detection** pipeline over 650K+ URLs, engineering 21 behavioral indicators such as IP presence, URL length, and suspicious keywords to expose phishing and malware patterns.
- Trained and compared Gaussian Naive Bayes, Random Forest, LSTM, Bi-LSTM, GRU, and CNN classifiers in Python, reaching 96%+ **detection accuracy** and generating actionable phishing and malware trend insights.

LST Capitals **November 2022 – January 2023**
Web Developer Intern *India*

- Developed Java **Spring Boot** REST APIs for bookings, pricing, coupons, and payments with Firebase real-time sync, sustaining sub-200ms latency across 50K+ booking records and 40% faster confirmations.
- Automated payment webhooks with **idempotent** handling for refunds, disputes, and chargebacks, and used transactional locking to prevent double-booked time slots, cutting manual refund handling by about 70%.

Skills

Languages: Java, Python, SQL, TypeScript, JavaScript, C++, Shell
Backend & APIs: Spring Boot, REST APIs, Microservices, FastAPI, Flask, Node.js, Maven
Databases: PostgreSQL, MySQL, SQL Server, Schema Design, Query Optimization, Indexing
Cloud & DevOps: AWS (S3, RDS, ECS/Fargate, Glue, CodePipeline), Docker, GitHub Actions, Jenkins, CI/CD, Git, Linux
Testing & Practices: Unit Testing, Integration Testing, PyTest, Mockito, JUnit, Code Reviews, Debugging, Agile/Scrum
Fundamentals & Security: Data Structures, Algorithms, OOP, Design Patterns, Malware Detection, Feature Engineering, AI-Assisted Development

Projects

Software Artifact Risk Scanner | *Java, Spring Boot, PostgreSQL, AWS S3, SQS, Docker, JUnit*

- Designed a Java **Spring Boot** REST service that queues uploaded build artifacts through S3 and SQS, extracts static risk features, scores them against policy rules, and stores auditable verdicts in **PostgreSQL**.

Chat2SQL Code | *Java, TypeScript, PostgreSQL, MySQL, Docker, Maven*

- Extended a **Java** and TypeScript database workspace built on open-source Chat2DB, supporting 10+ databases and generating schema-aware SQL with 82% accuracy using Docker, Maven, and Yarn builds.

Linguistix | *Python, Docker, AWS ECS/Fargate, CodePipeline, CI/CD*

- Developed and deployed a containerized Python **REST** data processing API on AWS ECS/Fargate with automated CodePipeline CI/CD releases and validation checks, achieving 87% accuracy and production-grade reliability.